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Automation Testing

I. Definition:

- Automation testing is the process of automatically run tests on a software application using special software tools.
- Instead of manually checking if the software works as expected, these tools do it for you, which saves time and reduces human error.
- The main goal is to ensure that the software behaves correctly by repeating tests quickly and efficiently, especially when changes are made.
- Automation testing needs manual effort when creating initial scripts, and further process is performed automatically to compare the actual testing result with expected results.

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II. Why Do We Do Automation Testing and Advantages of Automation Testing?

- We do automation testing to save time and effort for testing the application.

- With the help of automation testing, we have performed repetitive task which is monotonous by manually.
- Reduces manual work for testers.
- Finds bugs and errors quickly.
- Provides consistent and reliable results.
- Can run tests anytime.
- Runs many tests at once.
- Checks big and complex software easily.
- Reduces human error.
- Saves money in the long run.
- Tests on different devices and environments.
- Runs tests repeatedly as needed.

III. Challenges in Automation Testing?

- Writing good test programs takes time.
- Initial setup needs manual effort.
- Test programs can be complex to create.
- Test cases need to be updated with software changes.
- Not all tests can be automated.
- Requires skills and knowledge to write tests.
- Test programs can have bugs too.

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IV. Automation testing vs Manual Testing

Sr. No.	Automation Testing	Manual Testing
1	Uses special programs to test software.	Uses humans to test software.

2	Runs tests automatically without human help.	Tests are run by professional tester, not automatically.
3	Needs initial setup but then runs by itself.	Needs human effort for every test.
4	Finds bugs and errors quickly.	Can find bugs, but slower than automation.
5	Saves time compared to manual testing.	Takes more time compared to automation.
6	Can run tests many times easily.	Harder to repeat tests many times.
7	Checks big and complex software easily.	Checking big software can be tiring.
8	Reduces chance of human error.	More human error can be there.
9	Can run tests anytime.	Can only run tests during working hours.
10	Runs many tests at once.	Running many tests is time-consuming.
11	Improves software quality.	Improves software quality, but slower.
12	Requires skills to write test scripts.	Requires less technical skill initially.
13	Can be expensive to set up.	Cheaper to start with.
14	Handles repetitive tasks well.	Handles unique tasks well.
15	Test programs can have bugs too.	Human testers can miss bugs.
16	Requires good planning and design.	Planning and design are simpler.
17	Ensures coverage of all important parts.	Coverage depends on human testers.
18	Automates repetitive and time-consuming tasks.	Best for tasks that need human judgment.

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